SemEval-2026 Task 2: Predicting Variation in Emotional Valence and Arousal over Time from Ecological Essays

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Abstract

Longitudinal modeling of affect from text requires capturing both linguistic content and temporal emotional dynamics. SemEval 2026 Task 2 introduces a dataset of ecological essays and feeling words annotated with self-reported valence and arousal scores, enabling the study of affect as a time-evolving signal. In this work, we propose a neural architecture that combines pretrained Transformer encoders with temporal sequence modeling to predict continuous valence and arousal over user-specific timelines. Individual texts are encoded using a Transformer-based language model and aggregated through attention-based pooling before being processed by recurrent layers to capture longitudinal dependencies. To adapt pretrained representations under limited data conditions, we explore parameter-efficient fine-tuning strategies. We make the code available at `https://github.com/AndreaLolli2912/SemEval2026-EmoVA`.

1 Introduction

The affective circumplex model proposes that emotions can be described in a two-dimensional space defined by *valence* and *arousal*(Russell, 1980). While most affective computing research focuses on static snapshots or external perception of emotion, real emotional experience is inherently *longitudinal*—it evolves over time, shaped by daily routines, life events, and individual patterns.

Attempts have been made to address continuous affect prediction using pre-trained Tranformer architecture (Mendes and Martins, 2023) and to model longitude texts via hierarchical architecture with fixed-size representations and recurrent neural network (Christ et al., 2022), effectively mapping discrete emotion label to continuous space remains a challenge. Attention-based architecture, such **as Set Tranformers**, has been proposed to process set-structured data (Lee et al., 2019), reducing the operational cost from quadratic to linear, but implementation still requires careful design. SemEval-2026 Task 2 addresses this gap by introducing a longitudinal dataset of ecological essays and feeling words collected from U.S. service-industry workers over multiple years (2021–2024). This work focuses on **Subtask 1: Longitudinal Affect Assessment** and **Subtask 2a: Forecasting (future) Variation in Affect**. In this paper, we propose a hierarchical architecture combining `BERT`, Set Tranformer attention(ISAB/PMA) and bidirectional LSTM for longitudinal affect modeling. We show that the proposed solution achieves remarkable results, and that it provides satisfactory performance on small datasets by employing parameter-efficient fine-tuning strategies (LoRA, BitFit). To address the inherit valence-arousal difficulty gap, we design an asymmetric hybrid loss by combining Mean Squared Error (MSE) and Concordance Correlation Coefficient (CCC).

The rest of the paper is organized as follows: Section 2 introduces the task and datasets. Section 3 presents the main methodology adopted, with an overview of the pipeline, data preprocessing, and optimization objectives. Section 4 presents the main results obtained. Section 5 wraps up the paper, with considerations and possible future extensions of the work.

RC: alla fine dell'intrduzione metterei una versione condensata del background. aggiungere anche una piccola sezione con la descrizione di come è organizzato il resto del paper

2 Problem Description

RC: armonizzare la descrizione della problem description come fatto in paper reference: https://aclanthology.org/2025.semeval-1.287.pdf

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The goal of the task is to correctly predict current or future valence and arousal given a collection of texts. Two different datasets are provided: training data (2,765 samples) and test data (1,738 samples and 47 samples). Samples are collected as part of the "Data Science and Alcohol Consumption Study". Each sample is provided with different attributes and the correct valence/arousal values are provided for training data. Users contribute varying number of texts ranging from 2 to over 200 (Fig. 1), and discrete labels (Fig. 2).

Subtask 1 (ST1) requires predicting continuous valence and arousal scores for each essay, independently for each user. In Subtask 2a (ST2a), essays are provided together with preceding segments to model short-term changes in a user's emotional state. In Subtask 2b, the full history up to a split point is used to predict long-term dispositional changes in valence and arousal, capturing temporal evolution rather than absolute affective levels.

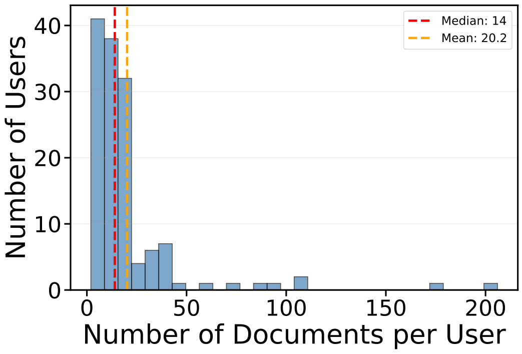

Figure 1: Number of documents per user in train data.

3 Proposed Methodology

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We hypothesize that affective states exhibit a temporal duality. Valence reflects rapid, reactive variations triggered by immediate events and explicit emotional language. Arousal, in contrast, follows a slower, inertial dynamic influenced by routines, persistent stress, and general well-being. Taking this into account, the proposed solution consists in one enconcoder-only pretrained Transormer, combined with attention-based aggregation and recurrent neural network (LSTM).

3.1 Data Preprocessing

Efficiently processing longitudinal data requires addressing the variability in both the number of documents per user and the length of individual texts. To enable parallelized batch processing, we implement a hierarchical padding strategy that standardizes input dimensions into a (B, S, L) tensor, where B is the batch size, S is the maximum number of documents per user, and L is the maximum token length.

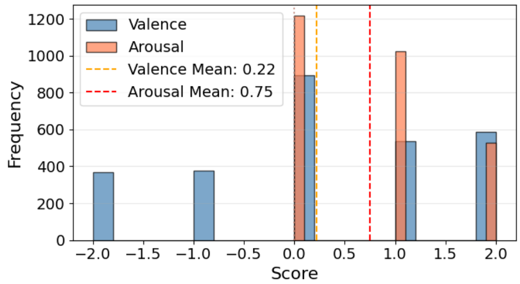

Figure 2: Distribution of Valence and Arousal scores in the training data, highlighting the contrast between the balanced valence labels and the skewed arousal labels.

Hierarchical Masking We generate a binary sequence mask $M_{seq} \in \{0, 1\}^{B \times S}$ to distinguish between genuine user texts and padding. This mask is propagated throughout the network to ensure that padded positions do not contribute to attention computations or the loss function.

Token Truncation At the document level, we utilize the `bert-base-uncased` tokenizer. Due to the quadratic complexity of attention mechanisms and GPU memory constraints, we limit the maximum token length to $L = 128$. This threshold balances semantic coverage with the memory efficiency required for processing multiple documents per user.

3.2 Model Architecture

Our architecture is a hierarchical neural pipeline that transforms longitudinal text into continuous valence and arousal predictions. It comprises four modules: (1) a Transformer encoder, (2) attention-based aggregation, (3) temporal modeling, and (4) a prediction head.

3.2.1 Encoder Backbone

We employ `bert-base-uncased` as the semantic encoder. Given an input of L tokens, the encoder outputs contextualized embeddings $H_{enc} \in \mathbb{R}^{L \times d}$, where $d = 768$. To reduce computational cost and prevent overfitting, we employ Parameter-Efficient Fine-Tuning (PEFT) strategies described in Section 3.4.

3.2.2 Attention-based Aggregation (ISAB & PMA)

To aggregate the variable-length sequence H_{enc} into a fixed-size document vector, we reject simple mean pooling in favor of the Set Transformer framework (Lee et al., 2019). This allows the model to learn which tokens carry affect ("signal") versus neutral context ("noise").

Multihead Attention Block (MAB) The fundamental building block is a variant of the Transformer block without positional encodings. Given a Query set Q and Key-Value set K, we define:

$$\mathrm{MAB}(Q, K) = \mathrm{LayerNorm}(H + \mathrm{FFN}(H)) \quad (1)$$

where the intermediate attention output H is:

$$H = \mathrm{LayerNorm}(Q + \mathrm{Multihead}(Q, K, K)) \quad (2)$$

Here, FFN denotes a standard row-wise feed-forward network with GELU activations. We fix the number of attention heads to $h = 8$ across all modules.

Induced Set Attention Block (ISAB) Standard self-attention scales quadratically $O(L^2)$. To efficiently filter the input tokens, we use ISAB, which approximates the attention mechanism using m learnable inducing points $I \in \mathbb{R}^{m \times d}$:

$$\mathrm{ISAB}_m(X) = \mathrm{MAB}(X, H) \in \mathbb{R}^{L \times d} \quad (3)$$

where the latent prototypes H are computed as:

$$H = \mathrm{MAB}(I, X) \in \mathbb{R}^{m \times d} \quad (4)$$

This reduces complexity to $O(Lm)$. The inducing points I first compress the input X into a lower-dimensional summary H, which is then used to contextualize the original input tokens.

Pooling by Multihead Attention (PMA) To compress the filtered sequence into a fixed-size representation, we apply PMA using a set of k learnable seed vectors $S \in \mathbb{R}^{k \times d}$:

$$V_{doc} = \mathrm{PMA}_k(X) = \mathrm{MAB}(S, X) \in \mathbb{R}^{k \times d} \quad (5)$$

In our configuration, we set $k = 8$. The resulting output is flattened into a vector $v_{doc} \in \mathbb{R}^{k \cdot d}$ to serve as the input for the temporal model.

3.2.3 Temporal Modeling and Inference

To capture the longitudinal dynamics of affect, the sequence of document embeddings is processed by an LSTM with a hidden dimension of 256. We fix the depth to 1 layer to mitigate overfitting. We treat directionality as a hyperparameter, evaluating both Unidirectional and Bidirectional configurations. The resulting temporal states are projected via two separate linear heads to predict valence and arousal respectively.

3.2.4 Multimodal Fusion

To address the forecasting objective of ST2, we adapted the temporal modeling architecture to integrate historical user data. We introduced two key modifications:

Multimodal Input Fusion: We employ a feature-level fusion strategy to combine textual and affective signals. At each timestep t, the aggregated document embedding is concatenated with the corresponding historical valence and arousal values. This enables the LSTM to jointly model the linguistic content and the user's affective trajectory.

State-Aware Prediction Head: The prediction mechanism is conditioned on the most recent observable state. We concatenate the final LSTM hidden state with the last known affective scores before the output projection. This design explicitly informs the model of the prior state, effectively simplifying the task to predicting the *variation* (or delta) required to reach the future state, rather than regressing the absolute values from scratch.

3.3 Optimization Objective

Predicting continuous affect presents a tension between two objectives: minimizing absolute error (magnitude accuracy) and preserving structural trends (correlation). A model trained solely on Mean Squared Error (MSE) tends to regress toward the mean, while pure correlation objectives can be numerically unstable.

The Zero-Variance Problem The gradient of correlation-based losses is inversely proportional to the prediction standard deviation $\sigma_{\hat{y}}$. In early training, if the model predicts near-constant values ($\sigma_{\hat{y}} \to 0$), gradients can explode, causing training instability.

Concordance Correlation Coefficient (CCC) To address this, we employ Lin's Concordance Correlation Coefficient (CCC), which measures both correlation and agreement in mean/variance:

$$\text{CCC} = \frac{2\sigma_{\hat{y}}\sigma_y\rho}{\sigma_{\hat{y}}^2 + \sigma_y^2 + (\mu_{\hat{y}} - \mu_y)^2} \quad (6)$$

By including the ground truth variance σ_y^2 in the denominator, CCC remains stable even when $\sigma_{\hat{y}} \approx 0$.

Combined Loss for ST1 Our loss function combines MSE for magnitude accuracy and CCC for structural alignment:

$$\mathcal{L}_{dim}(y, \hat{y}) = \lambda \cdot \mathcal{L}_{MSE}(y, \hat{y}) + (1-\lambda) \cdot (1 - \text{CCC}(y, \hat{y})) \quad (7)$$

The total loss is computed separately for valence and arousal:

$$\mathcal{L}_{total} = \omega_v \mathcal{L}_{dim}(v, \hat{v}) + (1 - \omega_v)\mathcal{L}_{dim}(a, \hat{a}) \quad (8)$$

We set $\lambda = 0.15$ (favoring CCC) and $\omega_v = 0.2$ (emphasizing arousal, which we found more challenging to predict).

3.4 Parameter-Efficient Fine-Tuning (PEFT)

Full fine-tuning of large language models on small longitudinal datasets often leads to overfitting. To address this, we freeze the majority of the encoder parameters and employ two efficient adaptation strategies.

Bias-term Fine-tuning (BitFit) Following Ben-Zaken et al. (2022), we freeze all attention and feed-forward weight matrices, training *only* the bias vectors. The trainable parameter set is defined as:

$$\Theta_{trainable} = \{b^{(l)} \mid l \in \text{Layers}\} \cup \Theta_{head} \quad (9)$$

where Θ_{head} denotes the task-specific parameters (ISAB, LSTM, Prediction Head). This approach drastically reduces the number of trainable parameters while allowing the model to shift activation distributions to align with the affect prediction task.

Weight-Decomposed Low-Rank Adaptation (DoRA) To capture more complex dependencies than bias updates allow, we employ DoRA (Liu et al., 2024). While standard LoRA approximates weight updates using low-rank matrices ($\Delta W = BA$), DoRA further decomposes the pre-trained weights into magnitude and direction components. This decomposition improves learning stability and capacity compared to standard LoRA. We apply DoRA to all linear layers with a rank of $r = 16$ and a scaling factor of $\alpha = 32$.

3.5 Implementation Details

The system is trained end-to-end using mixed-precision computation (FP16) to improve efficiency. Due to the memory overhead of the Set Transformer and LSTM unrolling, we use a micro-batch size of 1 with gradient accumulation to simulate an effective batch size of 16. Gradient clipping (1.0) is applied to stabilize optimization.

Differential Learning Rates When utilizing PEFT strategies, we apply a differential optimization schedule. We use a lower learning rate (5×10^{-6}) for the pre-trained encoder parameters to preserve linguistic knowledge, and a higher rate (1×10^{-4}) for the randomly initialized task-specific modules (ISAB, PMA, LSTM, prediction head). This prevents catastrophic forgetting in the backbone while allowing the new attention mechanisms to converge efficiently.

All hyperparameters, including loss weighting coefficients (λ, ω_v) and pooling configurations, are managed via experiment configuration files to ensure reproducibility.

4 Results

While training utilized the hybrid loss function described in Section 3 and MSE, model selection was performed using the official SemEval evaluation metrics on the validation set. For each dimension (Valence and Arousal), we compute the composite correlation by combining between-user and within-user Pearson correlations via Fisher's z-transformation. This composite correlation is the official ranking metric for the challenge. The overall score is the average of Valence and Arousal composite correlations. Results are summarized in Table 1 **RC**: fix table.

4.1 Overall Performance

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We present the main results obtained using the proposed pipeline, detailing the impact of various hyperparameter configurations on overall performance.

4.1.1 Result on ST1

The best performing model achieved a composite score of $r = 0.6802$ (Sim 10). This configuration combines a BERT-base-uncased encoder adapted via LoRA, an Induced Set Attention Block (ISAB)

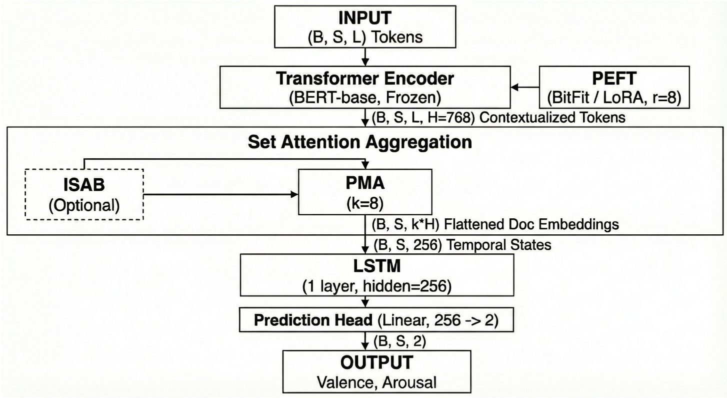

Figure 3: **RC**: TODO: si può rendere il diagramma più accattivante e usare colori? Overview of the proposed hierarchical architecture. The model processes longitudinal text sequences using a shared BERT encoder (optimized via DoRA), aggregates features via Induced Set Attention (ISAB) and Pooling by Multihead Attention (PMA), and models temporal dynamics with a Bidirectional LSTM.

with 32 inducing points, Pooling by Multihead Attention (PMA) with 8 seeds, and a unidirectional LSTM for temporal modeling. When ISAB and PMA are disabled, document representations are obtained via mean pooling over the Transformer's final layer token embeddings.

4.1.2 Impact of Attention Mechanisms

The introduction of ISAB and PMA provided significant performance gains over LSTM-only baselines. PMA alone (Sim 4, $r = 0.6211$) substantially outperformed the baseline without attention mechanisms (Sim 1, $r = 0.5776$). Adding ISAB further improved results: with 16 inducing points (Sim 6) the score increased to $r = 0.6555$, and with 32 inducing points (Sim 7) it reached $r = 0.6697$. Notably, PMA primarily improved Valence correlation, while ISAB corresponded to improvements in Arousal correlation. We hypothesize that ISAB filters noise in the user's textual history by attending to the most salient posts, which is particularly beneficial for the more challenging Arousal dimension.

4.1.3 Parameter-Efficient Fine-Tuning

We compared two PEFT strategies: LoRA and BitFit. LoRA (Sim 10, $r = 0.6802$) outperformed both BitFit (Sim 9, $r = 0.6640$) and the baseline without PEFT (Sim 7, $r = 0.6697$). The improvement from LoRA was primarily driven by Valence correlation ($r = 0.8049$), suggesting that adapting encoder weights is necessary to capture semantic nuances of emotional polarity. Arousal correlation also improved ($r = 0.5556$), though to a lesser extent. BitFit, while competitive, proved insufficient for optimal domain adaptation. Combining both strategies (Sim 11, $r = 0.6636$) caused a slight performance decline compared to LoRA alone, suggesting interference between the two adaptation methods.

4.1.4 Temporal Modeling

The effect of LSTM bidirectionality depended on architectural complexity. In simpler configurations without ISAB, bidirectional LSTMs provided improvements: Sim 3 (Bi-LSTM, $r = 0.6340$) outperformed Sim 4 (Uni-LSTM, $r = 0.6211$). However, once ISAB was introduced, unidirectional LSTMs performed better: Sim 6 (Uni-LSTM, $r = 0.6555$) outperformed Sim 5 (Bi-LSTM, $r = 0.6438$), and similarly Sim 7 ($r = 0.6697$) outperformed Sim 8 ($r = 0.6689$). We attribute this to functional redundancy. Both ISAB and bidirectional LSTMs model long-range dependencies; when ISAB handles global attention, the LSTM need only capture local sequential progression. Additionally, BERT already provides bidirectional contextualization at the token level, further reducing the benefit of a bidirectional recurrent layer. The unidirectional LSTM also has fewer parameters, mitigating overfitting on our small dataset.

4.1.5 Results on ST2

We evaluated different model configurations to identify the optimal strategy for forecasting. To this end, models were trained minimizing the **Mean Squared Error (MSE)** loss, while the **Pearson Correlation Coefficient** (r) was employed as the primary evaluation metric. As a baseline, we employed a **History-Aware Model** which utilizes the previous valence and arousal values concatenated

#	PEFT	ISAB	PMA	Bi-LSTM	Valence (r)	Arousal (r)	Overall (r)
ST1: Longitudinal Affect Assessment — Composite correlation ($r_{\text{composite}}$)							
1	–	–	–	No	0.6896	0.4655	0.5776
2	–	–	–	Yes	0.6849	0.4733	0.5791
3	–	–	8	No	0.7898	0.4525	0.6211
4	–	–	8	Yes	0.7608	0.5071	0.6340
5	–	16	8	No	0.7961	0.5150	0.6555
6	–	16	8	Yes	0.7683	0.5193	0.6438
7	–	32	8	No	0.7950	0.5444	0.6697
8	–	32	8	Yes	0.7835	0.5543	0.6689
9	BitFit	32	8	No	**0.8068**	0.5211	0.6640
10	LoRA	32	8	No	0.8049	**0.5556**	**0.6802**
11	Both	32	8	No	0.7880	0.5392	0.6636
ST2a: Forecasting Variation in Affect — Per-user Pearson (r_{user})							
1	–	–	–	No	0.4873	0.7579	0.6226
2	–	–	8	No	0.4443	0.6598	0.5521
3	–	–	8	Yes	0.4613	0.7606	0.6109
4	–	32	8	No	0.4994	0.5943	0.5468
5	BitFit	32	8	No	0.4576	**0.7790**	0.6183
6	BitFit	32	8	Yes	0.4199	0.7109	0.5654
7	LoRA	–	8	Yes	0.5444	0.6673	0.6058
8	LoRA	32	8	No	0.5257	0.7307	0.6282
9	LoRA	32	8	Yes	**0.6195**	0.6782	**0.6488**

Table 1: Ablation results on the validation set for ST1 and ST2a. Metrics shown are $r_{\text{composite}}$ (ST1) and per-user Pearson r (ST2a). ISAB and PMA columns indicate inducing points and seeds. **Bold** marks the best per metric within each task.

with the BERT embeddings, but lacks the specialized attention mechanisms (ISAB, PMA) and adaptation layers (LoRA) of our full architecture. The best performing model (sim. 9) achieved an average Score of **0.6488**, utilizing the winning architecture from ST1 augmented with a Bidirectional LSTM. Comparing our best model to the baseline, we observe that the introduction of historical data is not sufficient for balance performance since it fails to understand the Arousal trend. Regarding the best configuration, the introduction of **bidirectionality** led to a substantial improvement in **Valence** prediction ($+0.09$) compared to the unidirectional counterpart, effectively balancing the model's performance. Although the Arousal prediction experienced a slight decline, the overall composite score improved due to the better alignment of the valence trajectory. The previous findings regarding the necessity of Parameter-Efficient Fine-Tuning (LoRA) for semantic encoding and the capacity of Attention Mechanisms (ISAB and PMA) are confirmed also in the forecasting setting.

5 Discussion and Conclusion

We presented a hierarchical architecture for longitudinal affect assessment that combines BERT encoding, Set Transformer attention (ISAB/PMA), and recurrent temporal modeling. Our best configuration achieved a composite correlation of $r = 0.6802$ on the validation set for Task 1 and of $r = 0.6488$ for Task 2. Three factors proved critical for performance: (1) attention-based pooling (PMA) over mean pooling for document aggregation, with ISAB providing additional gains by filtering noisy historical context; (2) LoRA adaptation of the encoder, which outperformed both frozen backbones and BitFit; (3) unidirectional LSTMs when combined with ISAB, avoiding functional redundancy with the attention mechanism.

5.1 Limitations

Our approach has several limitations. First, the model treats each document as a flat token sequence, ignoring internal structure such as paragraphs or discourse segments. Second, the encoder relies solely on contextual embeddings without explicit affective knowledge. Third, evaluation was conducted only on the validation split; generalization to the official test set remains to be verified.

5.2 Future Work

Two directions appear promising for extending this work. Incorporating external affective lexicons (e.g., NRC, LIWC) as auxiliary features or through lexicon-guided pretraining could provide explicit emotional grounding. This may improve robustness, particularly for arousal prediction where contextual cues alone proved insufficient. Splitting documents into paragraphs or sentences and applying a second level of attention could capture intra-document structure. This hierarchical approach would allow the model to first aggregate sentence-level affect signals, then combine them into document representations, potentially improving sensitivity to localized emotional expressions.

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